

Article

A BLE 5.0 Extended Advertising Backscatter with Commodity Devices in Passive IoT Scenarios

Xiaoming Li *, Yuan Yuan, Yabin An and Bin Jiang

School of Microelectronics, Xidian University, Xi'an 710071, China; 21111212870@stu.xidian.edu.cn (Y.Y.); 19111110123@stu.xidian.edu.cn (Y.A.); 21111213006@stu.xidian.edu.cn (B.J.)

* Correspondence: lxm@mail.xidian.edu.cn

Abstract: BLE-based (Bluetooth Low Energy-based) backscatter has received considerable attention, as it aims to communicate with everyday smart devices such as smartphones, smartwatches, and tablets in passive IoT. The state-of-the-art BLE backscatter systems enable communication using a specialized continuous wave (CW) generator or entirely using commodity BLE 4.0 radios as an RF source. However, the existing BLE communication systems suffer from several key issues, including a short carrier length and a large frequency shift. This paper presents a passive BLE (PBLE) backscatter communication system that utilizes commodity BLE 5.0 radios. The system uses a BLE 5.0 extended advertising packet with partial single tones as excitations transmitting on the secondary advertising channel of BLE 5.0, and the BLE backscatter tag produces bandpass frequency-shift keying modulation at 1 Mb/s, which enables compatibility with BLE advertising channels. The prototype is implemented using an NRF52832 BLE 5.0 commodity chip, smart devices, and tags with FPGAs and chips. In FPGA board-level verification, when the downlink distance is 0.5 m, the uplink distance can reach 10 m. In chip testing, the uplink distance can reach 7 m when the downlink distance is 1 m. The baseband power consumption is 2 μ W, with a total power consumption of 10 μ W. This system eliminates the need for expensive and costly specialized RF sources, unlike the BLE backscatter communication system that uses a specialized CW generator. Compared to the BLE backscatter communication system that uses commodity BLE 4.0 radios, this system reduces the minimum frequency shift from 24 MHz to 2 MHz and increases the length of the single tones as a CW by a factor of about seven, from 31 bytes to 254 bytes.

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Keywords: passive IoT; backscatter; BLE 5.0; extended advertising packet; FSK

1. Introduction

In recent years, academia and industry have been investigating and researching the use of commodity devices and backscatter technology in conjunction with the signals of already-existing environmental devices such as Wi-Fi [1–9], LoRa [10], and BLE (Bluetooth Low Energy) [11–17] to conduct ‘scattering symbiotic communication’. The BLE communication protocol is currently supported by billions of cell phones, tablets and other smart devices [18–21]. At the same time, the BLE communication protocol specifies simpler physical and link layers. Therefore, the Bluetooth protocol is the most suitable communication protocol for backscatter technology compatible with commercial devices.

Various BLE backscatter systems have been proposed and developed recently [11–17]. MBS (modulated backscatter) is presented in [11] to verify the feasibility of BLE scattering. It utilizes an Agilent 33500B Arbitrary Waveform Generator (AWG) as an RF source. Ref. [12] introduces a method for producing BLE backscatter signals using an Agilent N5181A Signal generator to transmit a continuous wave (CW). Neither of the above systems provides a backscatter solution that is entirely built by commodity BLE radios. Both of

them require a specialized RF source, which is custom-built, expensive to deploy, and large in size, making them difficult to promote for certain applications.

Backscatter communication with commodity BLE 4.0 radios has received significant attention since specialized hardware is no longer needed. Through the use of Interscatter, [13] shows for the first time that Bluetooth radios can be used to create single-tone transmissions, but it is not compatible with BLE receivers. Through FS-backscatter, [14] uses a TI CC2650 BLE transmitter and a BLE receiver tuned to a channel that is 20 MHz away to achieve Bluetooth-to-Bluetooth backscatter. Freerider [15], RBLE [16], and IBLE [17] achieve backscattering using only commodity BLE 4.0 radios. All of the above BLE backscatter systems use BLE 4.0 radios to create single-tone transmissions. They eliminate the need for a specialized RF source, resulting in a significant reduction in cost and size. They leverage that Bluetooth uses Gauss frequency Shift Keying (GFSK), which encodes bits using two frequency tones. Thus, they transmit a stream of constant ones or zeros using reversely whitening techniques [13] to create a single-tone transmission as a CW. However, the above systems have some shortcomings.

1. **Short CW Length.** Since transmissions on data channels require the establishment of a connection with another device, the above BLE-based backscatter systems focus on Bluetooth advertisement channels where the systems can broadcast packets. Of the fields of a BLE 4.0 advertising packet, only the data payload can be set to arbitrary values. The length of the data payload ranges from 0 to 31 bytes. Therefore, the length of the constructed single tones using reversely whitening techniques as a carrier is no more than 31 bytes. Thus, the length of the whole regenerated packet is no more than 31 bytes, and the payload for modulating tag information is less than 15 bytes. The data rate of Bluetooth is 1 Mbps, so the maximum duration of the CW is 248 μ s. These limitations can negatively impact the systems' propagation capability;
2. **Large Frequency Shift.** Using a BLE 4.0 exciting advertising packet with partial single tones in the data payload field, the BLE-based backscatter system can backscatter it to another advertising packet in a different advertising channel. Bluetooth defines three advertising channels: 37, 38, and 39. The center frequencies for these channels are 2402, 2426, and 2480 MHz, respectively. The BLE specifications require the symbol 0 to be encoded as a negative frequency deviation 250 kHz below the channel center frequency and the symbol 1 as a positive 250 kHz deviation above the channel center frequency. The carrier frequency is shifted by different components to represent the symbols '0' and '1', respectively. When all zeros have a single tone in advertising channel 37 and the target channel is advertising channel 38, the needed frequency shift for direct frequency shift modulation is the minimum. The tag transmits information bit 0 with a minimum total shift frequency of 24 MHz, and information bit 1 with a minimum total shift frequency of 24.5 MHz;
3. **Complex BLE backscatter system design.** RBLE [16] can also backscatter an exciting advertising packet into a data packet. To modulate the symbol 0 to target channel 3, a frequency shift of 8 MHz is needed on the zero single tone of the exciting signal on advertising channel 37. A frequency shift of 8.5 MHz is needed for the symbol 1. Furthermore, RBLE [16] can be extended to work with BLE data packets. The data packet has a payload of up to 255 bytes. It reversely whitens BLE data packets so that the payload part of the data packet can be used as longer single tones to regenerate a BLE packet carrying more information. However, there are limitations to both approaches. This is because transmission on the data channel requires a connection to another device. After establishing a connection between two BLE devices, they communicate on the data channel to exchange data. Otherwise, the receiver cannot receive the backscattered BLE signal. Therefore, the implementation of the BLE-based backscatter system has become more complex, and the tag design requirements have also increased.

To overcome the above shortcomings, this paper introduces a passive Bluetooth (PBLE) backscatter system entirely utilizing commodity BLE 5.0 radios to generate single

tones as CW signals using reversely whitening in the secondary advertising channel of BLE 5.0. As shown in Figure 1, upon receiving single-tone signal from excitation BLE devices, the PBLE tag modulates sensor data onto them and backscatters new BLE advertising packet. The whole system involves only commodity BLE devices and a PBLE tag. Specifically, we make four key technical contributions to achieve this design.

4. We show for the first time that BLE 5.0 radios can be used to create a single-tone transmission. For Bluetooth using GFSK that encodes bits using two frequency tones we use reversely whitening techniques [13] to transmit a stream of constant ones or zeros to create a single-tone transmission (see Section 2.1.1);
5. A CW length of 254 bytes. The length of the constructed single tones using a BLE 4.0 advertising packet is no more than 31 bytes. We construct single tones over the advertising data field of a BLE 5.0 extended advertising packet which can be set to arbitrary values. The advertising data field can be up to 254 bytes in length. Therefore, the length of the constructed single tones using reversely whitening techniques as a carrier is extended up to 254 bytes. Thus, the length of the whole regenerated advertising packet can be 47 bytes, and the payload for modulating tag information is up to 31 bytes. The data rate of Bluetooth is 1 Mbps, so the maximum duration of the CW is 2032 μ s (see Section 2.1.2);
6. A smaller frequency shift. We leverage that BLE 5.0 also enables the transmission of an extended advertising packet on the 37 data channels in addition to the advertising channels 37, 38, and 39. Thus, to modulate a symbol 0 to target advertising channel 39, we only need to shift a frequency of 2 MHz on the zero single tone of the exciting signal on data channel 36. A frequency shift of 2.5 MHz is used for the symbol 1. Compared to the 24 M and 24.5 M frequency shifts in the state-of-the-art BLE backscatter communication systems that use only commodity BLE 4.0 radios, the 2 M and 2.5 MHz frequency shifts significantly reduce the power consumption of the tag. (see Section 2.2);
7. An easier-to-implement BLE-based backscatter system. We backscatter an exciting BLE 5.0 extended advertising packet with partial single tones into a new BLE advertising packet. Since the transmission of advertising packets does not require the establishment of a connection with another BLE device, we only focus on broadcasting a BLE 5.0 extended advertising packet. As a result, we reduce the complexity of the BLE-based backscatter system implementation and simplify the design of the tag while reducing the frequency shifts to 2 MHz and 2.5 MHz and increasing the single-tone length to 254 bytes compared to RBLE [16] (see Section 2.4).

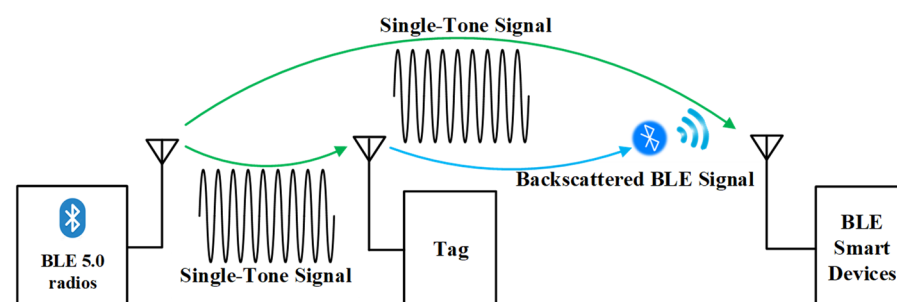


Figure 1. Conceptual design of PBLE entirely using commodity BLE 5.0 radios. The PBLE tag modulates its sensor data on BLE 5.0 single-tone signal and backscatters new BLE advertising packets that any commodity BLE device can decode.

We conduct experiments using FPGA board-level tags and chip tags. Our evaluation shows we can generate a BLE advertising packet from the single-tone transmission of a BLE 5.0 extended advertising packet. The experiments indicate that the uplink communication distance R_2 , which is the distance between the tag and receiver, can reach up to 10 m when the downlink distance R_1 , which is the distance between the transmitter

and tag, is 0.5 m during FPGA board-level verification. In the chip test, the baseband power consumption is 2 μ W and the total power consumption is approximately 10 μ W. Compared to the total power consumption being 30 μ W for Freerider [15], 37 μ W for RBLE [16], and 31 μ W for IBLE [17], our work significantly reduces the power consumption of the tag chip.

2. System Design

BLE 5.0 has added extended advertising. Extended advertising, as a newly added feature, increases the payload amount of transmitted advertising packet data. In the BLE 4.0 specification, there are only three advertising channels: 37, 38, and 39. And the 0–36 channels are data channels. Advertising packet data can only be transmitted on an advertising channel, and the payload of an advertising packet is no more than 31 bytes, as shown in Figure 2. BLE 5.0 defines the three advertising channels 37, 38, and 39 as primary advertising channels, and the 0–36 data channels as secondary advertising channels [22]. Secondary advertising channels can also be used to transmit advertising data. This is the core idea of BLE 5.0 extended advertising: using data channels to transmit advertising data.

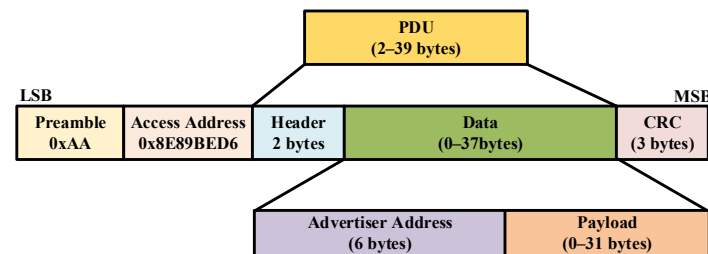


Figure 2. Structure of a BLE 4.0 advertising packet.

Figure 3 shows the Bluetooth channel diagram. We transmit extended advertising packets with the advertising data field of 254 bytes on the secondary advertising channel 36, which builds the 254-byte single tones of zeros and reduces the required frequency shift to modulate a symbol 0 and a symbol 1 for backscattering to 2 MHz and 2.5 MHz, respectively.

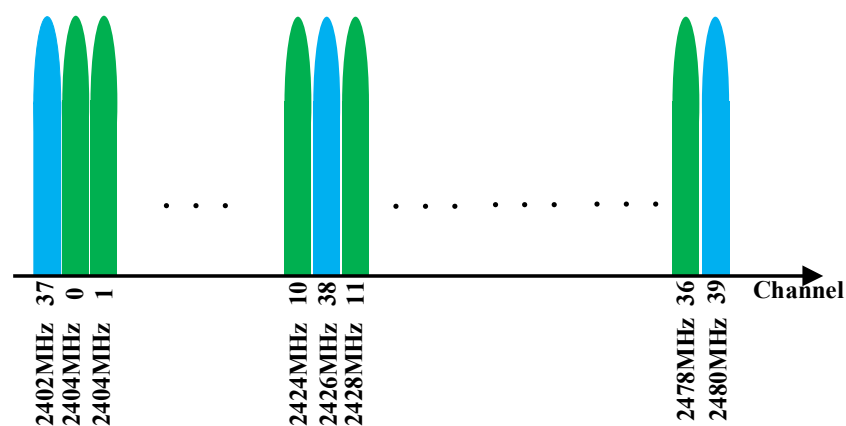


Figure 3. Bluetooth channel diagram. The channels colored in blue are advertising channels. The channels are colored in green are data channels, also named as the secondary advertising channel.

2.1. Commodity BLE 5.0 Radios as an RF Source

The use of current BLE general-purpose chips in mobile terminals for generating single tones as CW excitation, such as the Nordic BLE chip, eliminates the need for a specialized RF signal source, resulting in a significant reduction in cost and size. In the studies [13–17], a BLE chip is used to transmit single tones as a carrier for backscatter communication.

The objective is to transform the Bluetooth chip into a single-tone transmitter with constant amplitude and frequency. This can be achieved by utilizing two insights about the GFSK modulation used in Bluetooth. Firstly, Bluetooth uses two frequencies to encode zero and one data bit. Therefore, if a stream of constant ones or zeros can be transmitted, a single frequency tone can be created. The Gaussian filter utilized by Bluetooth does not alter the spectral properties of a single tone, as it only serves to smooth out abrupt frequency changes. Thus, transmitting a continuous stream of ones or zeros through the Bluetooth chipset can effectively generate a single tone. However, achieving this requires overcoming two main obstacles: data whitening and the BLE 5.0 extended advertising packet structure.

2.1.1. Data Whitening

While the goal is to create long sequences of zeros or ones, Bluetooth uses data whitening to avoid such sequences in order to achieve accurate timing recovery at the Bluetooth receiver. When long periods of ‘0’ and ‘1’ occur in the transmitted data, the receiving system will not show level changes for a long period of time, and the timing signal value obtained will be lower than the ambient noise, resulting in system interruptions and errors.

InterScatter [13] technology utilizes a reversely whitening technique to convert commercial BLE advertising signals into a CW, eliminating the need for specialized RF excitation sources. Specifically, Bluetooth uses the 7-bit linear feedback shift register circuit in Figure 4 with the polynomial shown in Equation (1). The circuit outputs a series of bits that are used to whiten the input data by performing an out-of-parallel operation on the data bits with the bits are output by the circuit, based on an initial state. We can invert this whitening process to create the desired sequence of ones or zeros. It is possible to deterministically generate the whitening sequence based on the initial state of each register. According to the Bluetooth specification, the shift registers are initialized using the Bluetooth channel number. The zeroth register is set to a one, and the binary sequence of channel numbers is set in the rest registers. For example, when transmitting on Bluetooth advertising channel 37, the zeroth register in Figure 4 is set to 1 and the remaining registers are set to the binary representation of 37. Thus, given an advertising channel, we can initialize the Bluetooth whitening algorithm and compute the whitening sequence. The data bits are then set to the same bits in the whitening sequence or their bit complements to generate a long sequence of zeros or ones, respectively. In Section 3, we show that this procedure works for unmodified Bluetooth chipsets.

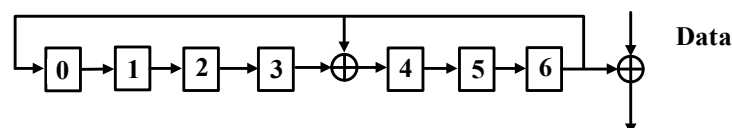


Figure 4. Whitening processing circuit structure.

$$\text{whiten} = x^7 + x^4 + 1 \quad (1)$$

2.1.2. BLE 5.0 Extended Advertising Packet Structure

This discussion assumes that all bits in a BLE 5.0 extended advertising packet can be controlled. However, an extended advertising packet has fields that cannot be arbitrarily modified. Figure 5 displays the PDU structure of the BLE 5.0 extended advertising packet. The PDU structure of an extended advertising packet consists of a two-byte header and a number of bytes of payload. The payload comprises the extended header length, advertising mode, extended header, and advertising data fields. Of these fields, only the advertising data field can be set to arbitrary values. The length of the advertising data ranges from 0 to 254 bytes. Therefore, the maximum length of the constructed single tones as a carrier is 254 bytes. Thus, we can regenerate the BLE advertising packet shown in Figure 2 with the 31-byte payload for modulating tag information. The data rate of Bluetooth is 1 Mbps, so the maximum duration of the CW is 2032 μ s.

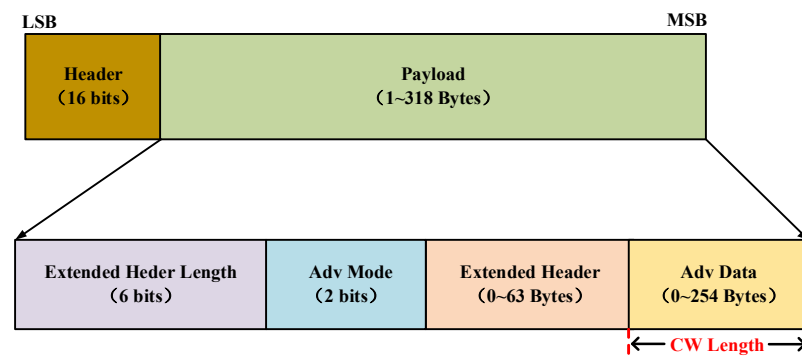


Figure 5. Structure of a BLE 5.0 advertising packet.

2.2. Modulation Using Direct Frequency Shift

In a BLE channel of 2 MHz, the minimum unilateral frequency deviation is 185 KHz and the maximum is 500 KHz. A positive frequency deviation above the channel center frequency of 185 KHz represents the '1' bit, while a negative frequency deviation below the channel center frequency of 185 KHz represents the '0' bit. According to reference [16], the lowest bit error rate (BER) for received data in backscatter communication systems is achieved with a frequency deviation of 250 KHz. Thus, as shown in Figure 6, we use a positive frequency deviation f_d , 250 KHz to represent the symbol 1, and a negative frequency deviation of the same amount is to represent the symbol 0, following GFSK.

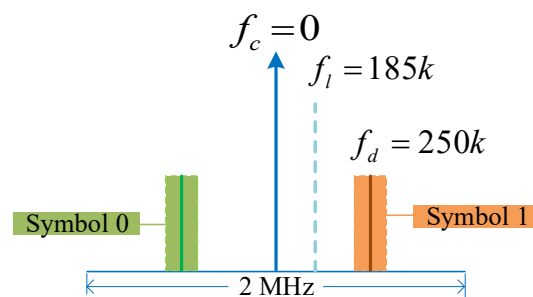


Figure 6. BLE GFSK modulation.

Inspired by Interscatter [13], by properly performing reversely whitening techniques, the advertising data field of the BLE 5.0 extended advertising signal can be made all ones or zeros, which are single tones. Next, we directly apply frequency shifts to modulate 0 or 1 in the target channel.

BLE 5.0 introduces eight types of extended advertising packets. Only the ADV_EXT_IND packet works on the primary advertising channel. The remaining seven

types of extended advertising packets work on the secondary advertising channel. To construct a single-tone signal as a carrier, we transmit extended advertising packets on the secondary advertising channel 36. As a result, we increase the single tone length to 254 bytes. And we reduce the required frequency shifts for backscattering to 2 M and 2.5 M.

As shown in Figure 7, we have an all-zeros single tone in the secondary advertising channel 36, and our target channel is the advertising channel 39. When we need to modulate a symbol 0, a frequency shift of 2 MHz, Δf_1 , is chosen to modulate. Similarly, a frequency shift of 2.5 MHz, Δf_2 , is able to modulate the symbol 1.

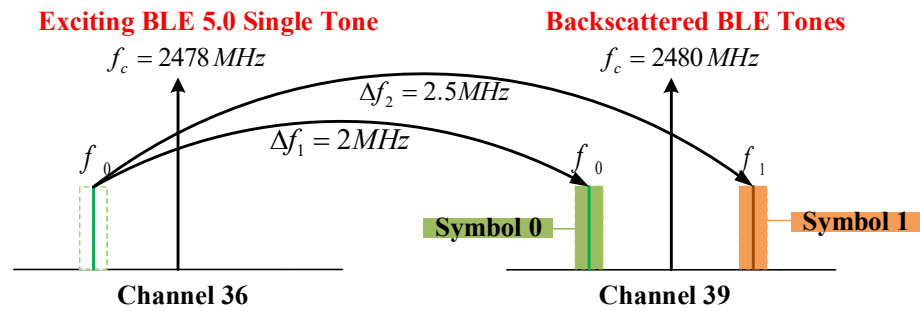


Figure 7. Direct frequency shift modulation.

2.3. Overall Flow of AUX_SYNC_IND Packet

We choose the AUX_SYNC_IND packet with extended advertising transmitting on the secondary advertising channel to construct the single tones as a carrier. The reason for this is that the AUX_SYNC_IND packet can be broadcasted cyclically at a fixed interval, just like the normal advertising packet. Two other types of extended advertising packets are required for the transmission of the AUX_SYNC_IND packet. These two types of extended advertising packets are ADV_EXT_IND and AUX_ADV_IND.

The broadcast sequence of the AUX_SYNC_IND packet is shown in Figure 8. The advertiser sequentially transmits the ADV_EXT_IND packet on the primary advertising channels 37, 38, and 39. The packet includes the data channel on which the AUX_ADV_IND packet is transmitted and the time at which the AUX_ADV_IND packet appears on the secondary advertising channel. This indicates when and where the receiving device can receive the AUX_ADV_IND packet. The AUX_ADV_IND packet provides the information about the first appearance of AUX_SYNC_IND, including its timing, the fixed interval, and the access address, etc. According to the information in the AUX_ADV_IND packet, the advertiser and the receiving device are synchronized, so both wake up at the same time. The advertiser transmits data, and the receiving device receives the data.

Therefore, we configure the ADV_EXT_IND and AUX_ADV_IND packets so that the transmitter periodically transmits the AUX_SYNC_IND packet. We use reverse whitening to construct a 254-byte single tone of zeros in the advertising data field of the AUX_SYNC_IND packet as a CW.

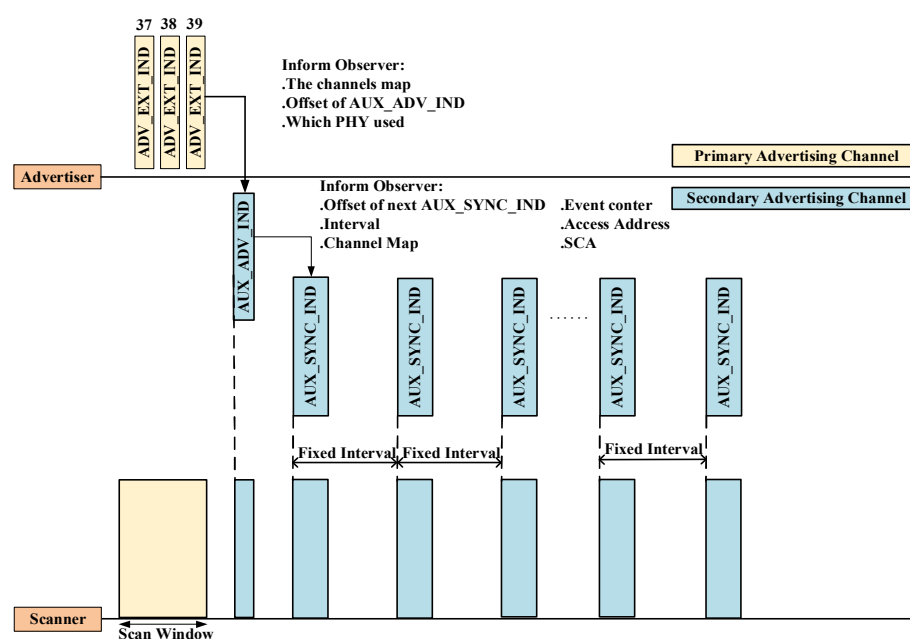


Figure 8. Flowchart of AUX_SYNC_IND packet advertising.

2.4. PBLE Backscatter System Excited with Commodity BLE 5.0 Radios

2.4.1. Conceptual Design

Figure 9 shows the commodity BLE device used as an excitation source to construct a prototype PBLE backscatter system, consisting of three parts: a commercial excitation source, a tag, and a smart device. The commercial excitation source must follow the anti-whitening techniques [13] to construct a single-tone signal in the advertising data field of the BLE 5.0 extended advertising packet. When a commercial excitation source generates a single-tone signal as a CW using reversely whitening techniques, the tag monitors the CW using averaging detection. After skipping a series of non-single-tone character fields in front of the advertising data field of the AUX_SYNC_IND packet, the internal data of the tag are backscattered through the antenna. At this point, the smart device receives the tag's backscattered signal, which complies with the BLE advertising standard [23].

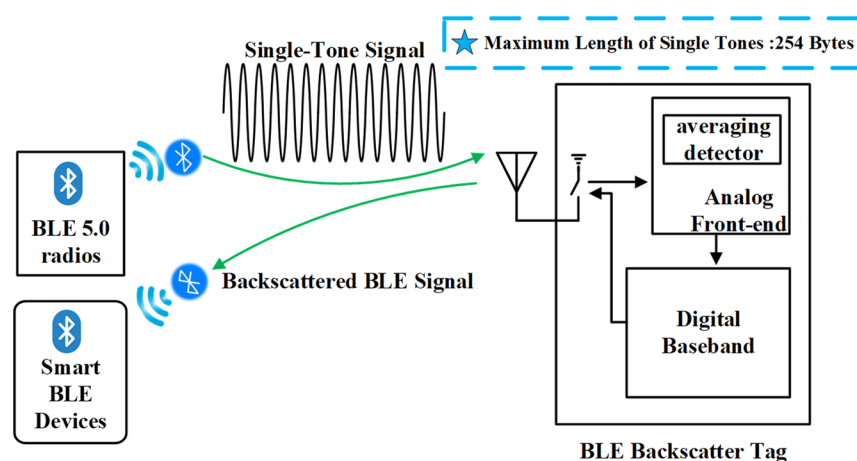


Figure 9. Prototype PBLE backscatter scheme for commercial BLE 5.0 device excitation.

2.4.2. Framework

Figure 10 shows the framework of the PBLE backscatter system excited with commodity BLE 5.0 radios.

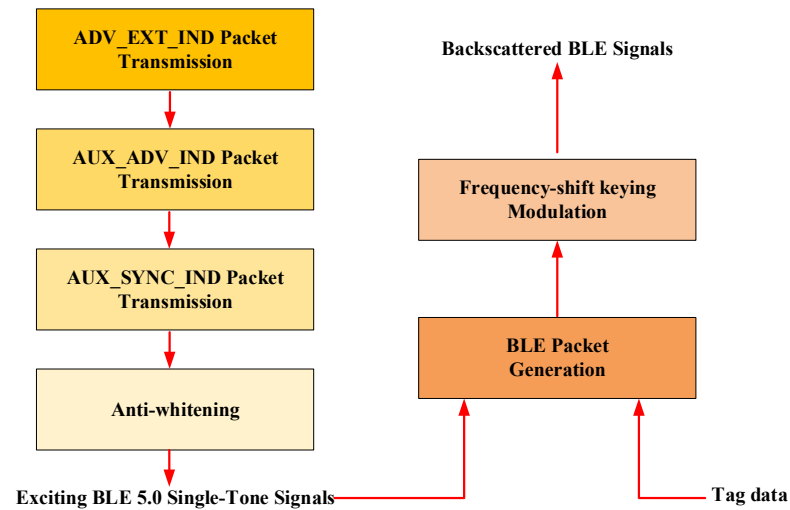


Figure 10. The framework of PBLE backscatter system excited with BLE 5.0 radios.

The implementation of a PBLE backscatter communication system based on commodity BLE 5.0 radios is as follows:

- The commodity BLE device uses the NRF52832 BLE chip to transmit the ADV_EXT_IND packet on the primary advertising channels 37, 38, and 39 in sequence. The data frames of the ADV_EXT_IND packet and AUX_ADV_IND packet are pre-set to carry the information of the AUX_SYNC_IND packet. The advertising channel is the target channel. A string of single-tone signals consisting of zeros is constructed in the advertising data field of the AUX_SYNC_IND packet on the secondary advertising channel 36. The signals are created in accordance with the anti-whitening techniques and serves as the backscatter carrier source for the PBLE tag. The carrier frequency is 2477.75 MHz. The target channel is labeled as the advertising channel 39. The total shifted frequencies are 2 and 2.5 MHz, respectively, as shown in Figure 7.
- The tag completes the generation of the BLE advertising data frames and outputs the advertising packet data serially.
- In the backscattered frequency shift keying (FSK) modulation method, the tag transmits information by selecting different shift frequencies for carrier modulation. The tag achieves this by using ultra-low-power backscatter modulation. To enable smart devices to receive backscatter data, the external carrier in the data channel is moved to the BLE advertising channel. Figure 11 shows that the tag's backscatter modulation has similar characteristics to the mixer. The mixer alters the reflection coefficient of the antenna's matching impedance by the baseband signal inside the chip. This allows for the multiplication of the external carrier and the internal signal, completing the FSK modulation and generating the BLE backscattered signal.

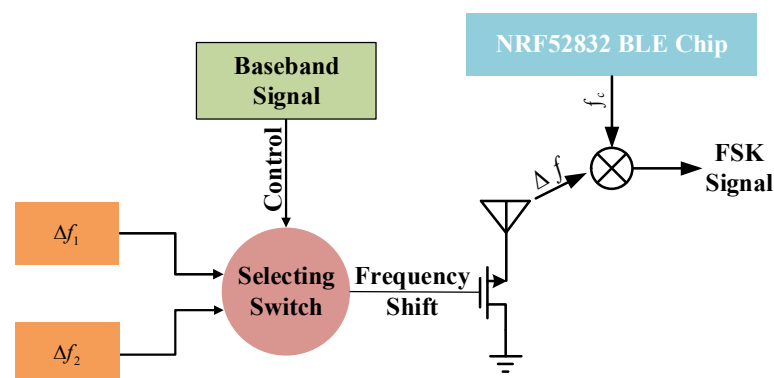


Figure 11. Block diagram of BLE 5.0 backscatter modulation.

3. Implementation

We make a prototype PBLE using an NRF82532 chip and a tag implemented by FPGAs to build proof-of-concept applications. We then translated the tag into an IC and used it to quantify the power consumption.

3.1. FPGA Tag Verification

Figure 12 shows the test scenario for the integrated board-level tag. The distance between the tag and the receiving device (iPad) is referred to as the uplink distance. The NRF82532 chip is powered by a 2.5 V battery and generates a single-tone carrier at a frequency of 2477.75 MHz with a transmitting power of -4 dBm. The board-level tags are powered by a rechargeable battery. The frequency shift of the board-level tags is 2 MHz and 2.5 MHz, generated by an internal phase-locked loop in the EP4CE6 series FPGAs. The data frame is used to control the selecting switch to determine the frequency shift. The modulator uses the ADG902 chip, which operates at a voltage of 2.5 V. By adjusting the frequency control of the reflective modulator loaded onto the single-tone carrier, the resulting upper sideband signals can be directed towards the BLE advertising channel 39. This enables the smart device to receive the backscattered BLE advertising signal.

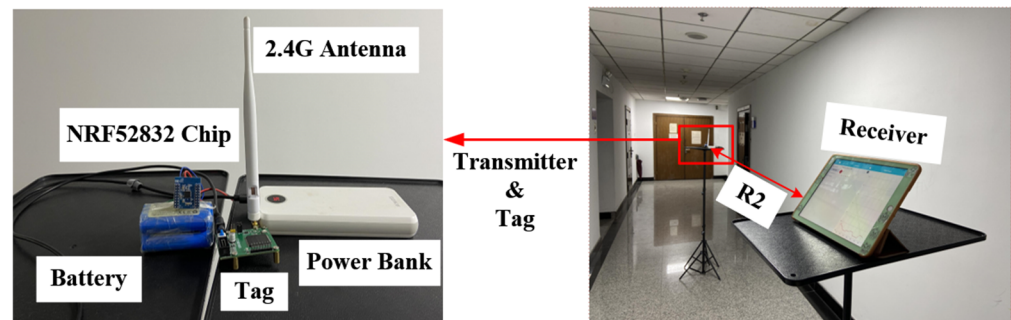


Figure 12. Construction of experimental test scenario.

Figure 13 displays the test results of the BLE 5.0 single-tone signal generated by the NRF52832 chip. The plot shows that we can create single-tone transmissions from commodity BLE 5.0 devices.

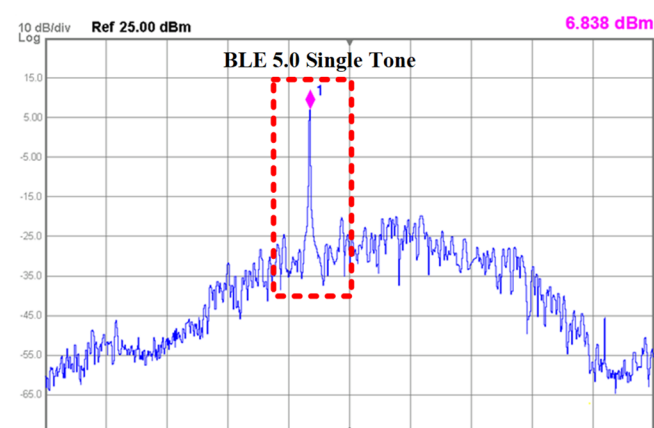


Figure 13. Spectrum of single-tone transmissions of zeros. The red dash square indicates the position of the single tone. The pin prismatic represents the marker point, and the number “1” represents the marker number by frequency spectrum analyzer.

Based on the advertising data generated above, the tag successfully transmits data to the smart device. Figure 14 shows the nRF connect software page of version 4.22.3 captured by the iPad. The iPad reads the advertising packet named ‘x’. On the right side

of Figure 14 is the received signal strength indication (RSSI) graph. The x-axis represents the moment of reception and the y-axis represents the RSSI value of the received BLE signal. This shows that our work can successfully regenerate new BLE advertising packets and achieve stable reception.

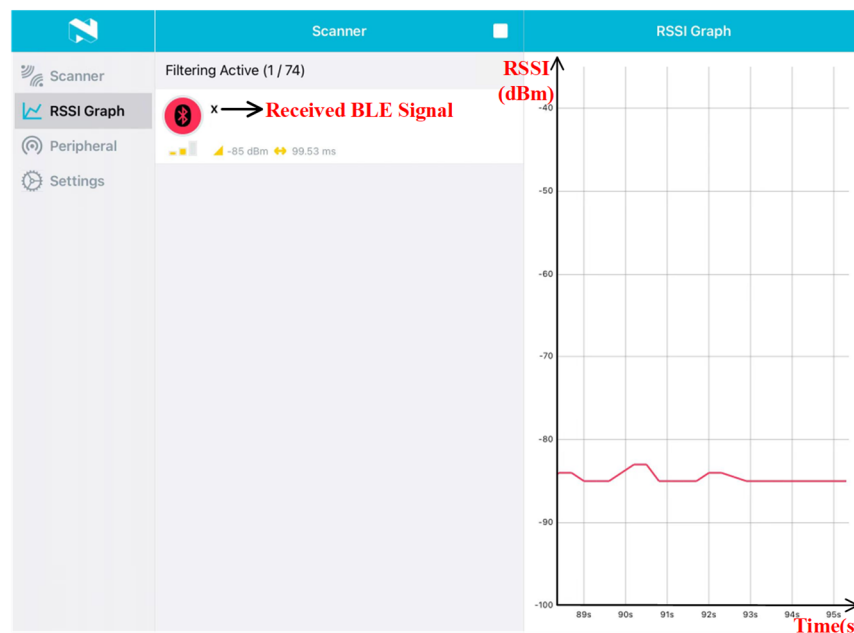


Figure 14. The BLE packet generated by the tag is received by an unmodified iPad.

The distance R1 between the transmitter and tag is shown in Figure 12. Through moving the receiver, the uplink communication distance R2 between the tag and receiver can reach up to 10 m. In Figure 15, the x-axis plots the distance R2 between the receiver and the tag while the y-axis plots the RSSI values of the received BLE signal. With the increase in R2, the intensity of the signal received by the receiver gradually weakens. The signal strength of the regenerated BLE advertising packets is strong within the uplink communication distance R2 of 3 m.

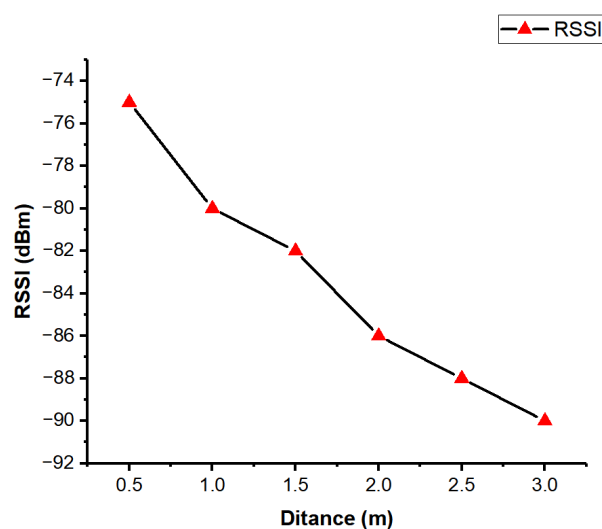


Figure 15. Backscatter RSSI changed with uplink distance.

3.2. IC Tag Verification

The circuits in this paper were finally produced and flowed in TSMC 180 nm standard CMOS process node. Figure 16 shows the chip test environment. The Agilent E4438C RF source powers the tag, the NRF82532 BLE module provides a single-tone CW carrier, and the tag generates serial signals of advertising packet data frames that need to be uploaded. Driven by the extended advertising data frame, the tag node successfully transmits data to the smart device. The iPad reads the BLE advertising packet with the name “XDU” using the nRF Connect software of version 4.22.3. When the downlink distance R1 is 1 m, the uplink distance R2 can be up to 7 m.

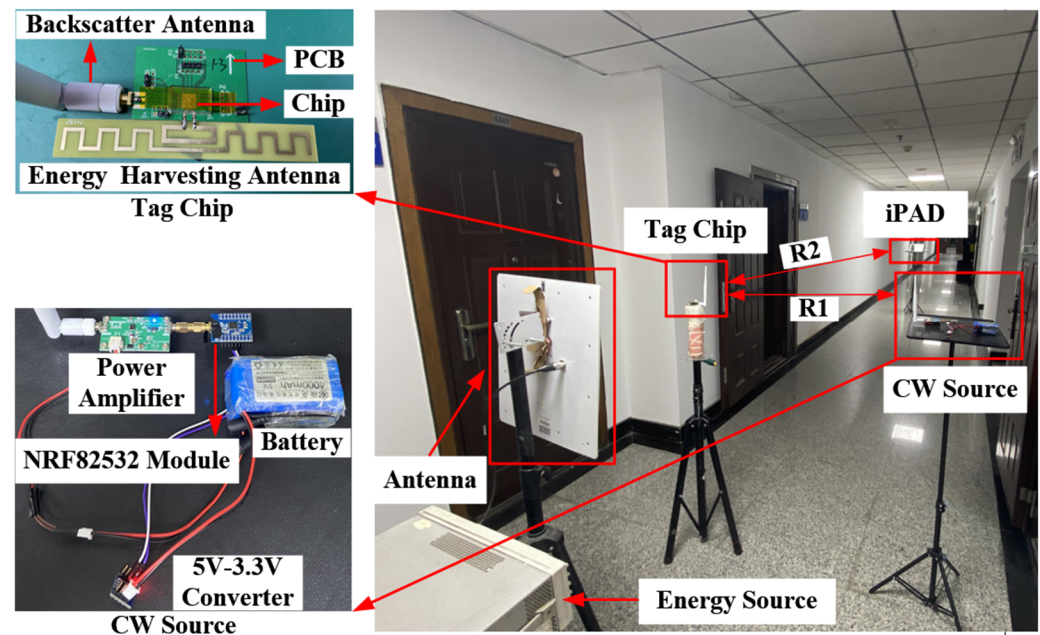


Figure 16. Chip test scenario diagram.

Figure 17a shows that the received BLE signal strength decreases as the uplink distance increases. At an uplink distance of 7 m, the received signal strength is -99 dBm. Figure 17b demonstrates that the iPad can reliably receive the Bluetooth signal generated by backscattering.

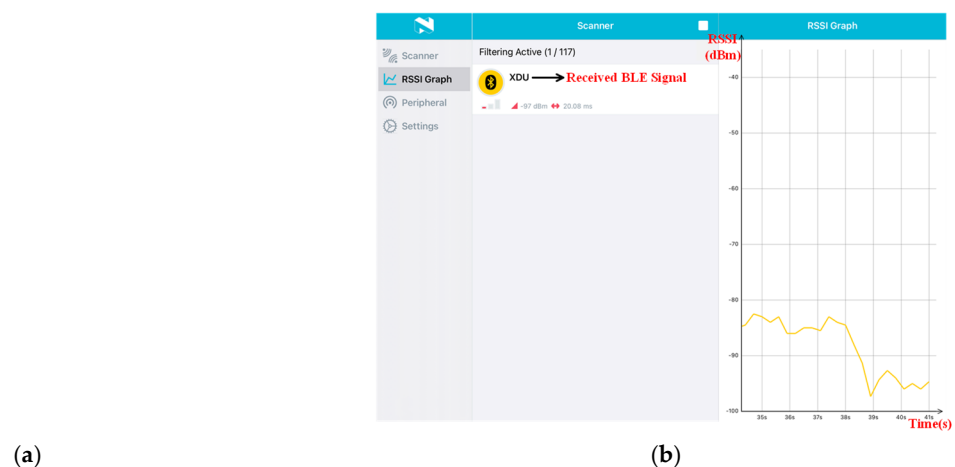


Figure 17. Result of IC design test. (a) Backscatter RSSI changes with uplink distance; (b) the BLE packet generated by the tag is received by an unmodified iPad.

Table 1 summarizes the performance of our work and compares it with the state of the art.

Table 1. Comparison of BLE backscatter system.

Reference	[11]	[12]	[13]	[14]	[15]	[16]	[17]	This Work
The overall tag power consumption (μW)	N/A	N/A	28	45	30	37	31	10
CW source implementation	Agilent 33500B AWG	Agilent N5181A Signal Generator	TI Bluetooth device	CC2650 BLE transmitter	TI CC2541	TI CC2540	TI CC2540	NRF52832
CW source type (Com: Commodity)	specialized CW generator	specialized CW generator	Com BLE 4.0 radios	Com BLE 4.0 radios	Com BLE 4.0 radios	Com BLE 4.0 radios	Com BLE 4.0 radios	Com BLE 5.0 radios
Maximum CW length (Bytes)	31	31	31	31	31	31	31	254
Duration of single tones (μs)	248	248	248	248	248	248	248	2032
Maximum receiving distance (m)	R1 + R2 = 9.4	R1 = 1 R2 = 30	R1 = 1 R2 = 27	R1 = 1 R2 = 4.8	N/A R2 = 12	R1 = 0.3 R2 = 25	R1 = 1.5 R2 = 20	R1 = 0.5 R2 = 10
Modulation type	FSK	FSK	PSK	FSK	FSK	BFSK	IPS/GFSK	FSK
Frequency shift (MHz)	25.5	12	11	20	20	8	8	2
FPGA/IC tech. (nm)	FPGA	FPGA	FPGA	FPGA	FPGA	65	65	180

By utilizing commodity BLE 5.0 radios to generate a single-tone signal as a CW in the advertising data field of a BLE 5.0 extended advertising packet, the maximum length of the signal can be extended up to 254 bytes. The BLE data rate is 1 Mbps, allowing for a single-tone duration of up to 2032 μs . This increases the performance and reliability of backscatter communications, as well as the amount of data that can be backscattered. The required frequency shift is only 2 MHz, and the total power consumption of the chip implementation is only 10 μW . Compared to IBLE [16] and RBLE [17], our work requires significantly less power for the passive tag chip to implement a BLE backscatter communication system, which is a key point for PBLE tag commercial implementation, like with today's passive RFID tag.

4. Conclusions

This paper proposes an advanced and reliable PBLE backscatter system that uses commodity BLE 5.0 radios as an RF source and works with a single smart device as a receiver. The main contributions lie in using BLE 5.0 extended broadcast signals with partial single tones as excitations for BLE backscatter. This solves the problem of traditional BLE backscatter requiring a specialized RF source. By utilizing a BLE 5.0 extended advertising packet with partial single tones as excitations, the length of the CW can be extended from 31 bytes to 254 bytes. Therefore, the payload of the regenerated BLE advertising packet for modulating tag information increases from 15 bytes to 31 bytes. As the BLE data rate is 1 Mbps, the duration of the single tones can be 2032 μs , largely increasing the success rate of backscatter communication. A string of single-tone signals consisting of zeros is created on data channel 36 using the reversely whitening technique. The baseband generates a BLE signal through backscatter on advertising channel 39. The smart device can then receive the BLE signal. In this process, the frequencies needed to construct bit 0 and bit 1 are 2 MHz and 2.5 MHz, respectively. By using the commodity BLE 5.0 radios to generate the single-tone signal as a carrier, the minimum frequency shift

has been reduced from 24 MHz to 2 MHz. This reduction in frequency shift significantly reduces the power consumption of the system. The test results indicate that in FPGA board-level verification, the uplink distance can reach up to 10 m when the downlink distance is 0.5 m. In chip testing, the baseband power consumption is 2 μ W, and the total power consumption is approximately 10 μ W. Additionally, the uplink distance can reach up to 7 m when the downlink distance is 1 m. This study clarifies the basics of constructing a Bluetooth backscatter system with commodity Bluetooth 5.0 radios. The proposed solution demonstrates practical feasibility and significantly reduces power consumption and deployment costs while improving backscattering reliability. This further promotes the use of BLE in backscatter systems.

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